CHAPTER 3

Brief History of DevOps and DevSecOps

In the first two chapters, we went over the new and updated material included in the second edition, including the following:

- Expanded coverage of DevOps/DevSecOps history
- Latest version of the *DevOps with LEGO and Chocolate* simulation
- Addition of the two new coaching games designed for enabling a culture of innovation and learning
- A robust collection of debriefing practices that will support you in facilitating group conversations with more impact
- Books and experiences that inspired me to start designing Agile/DevOps coaching games, embracing playfulness in my training and coaching

The following two chapters will offer you a few notable highlights of the DevOps/DevSecOps history and walk you through a variety of concepts these games are designed to teach. My goal is to offer you a DevOps

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primer, including some key theories, as well as important highlights from notable industry reports. Think of these chapters as "minimum viable basics"—just enough to grasp the ideas built into this gamified DevOps/DevSecOps introduction, and to be able to connect them with real-life practices during debriefing. This will be sufficient to get you started, and by no means do I expect you to stop here. Continue your own DevOps upskilling with a new edition of *The DevOps Handbook*,[1] and other great resources published by IT Revolution Press.

If you are already familiar with these ideas, feel free to skip directly to Chapter 5 and start diving into the dynamics and mechanics of the simulation game.

DevOps

What is DevOps? What kind of fundamental challenges created a fertile ground for it to emerge in the first place? What are the key principles underpinning DevOps? Where is it now, 14 years since the term "DevOps" was coined by Patrick Debois? What kind of technical DevOps practices are modeled in the *LEGO and Chocolate* simulation, and what do you need to know about them to successfully run this simulation?

Functional Silos and a Low-Trust Culture

It is hard to imagine now that less than fifteen years ago we didn't even have the term "DevOps" in our vocabulary. In fact, organizations around the world were mostly still at the beginning of their Agile journeys. According to the 2009 State of Agile Report by VersionOne, only 31% of

[1] Kim, G., Humble, J., Debois, P., Willis, J., Forsgren, N. (2021). *The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations*. IT Revolution Press.

respondents worked in organizations with "two teams or less" practicing Agile, while others were still applying the traditional Waterfall approach to managing and delivering projects.

While a change in the speed of delivery and collaboration patterns at least started to happen on the development side at that time, the processes, tools, and behaviors on the operations side were still largely unchanged. It was the time of large data centers, expensive relational databases, and physical servers that had to be ordered from hardware vendors three to nine months in advance. Building a new QA environment, loading production-like data, or deploying new code into production required long, error-prone manual efforts on the part of IT operations.

It was the time of monthly (or even quarterly) deployments. Handling post-production deployment emergencies, carrying pagers, and invoking escalation procedures were all parts of "business as usual" for IT operations. While many of these emergencies were caused by the release of new versions of applications (implemented earlier by the development teams), it was the on-call database administrator (DBA) or the on-call system administrator (SA) who was paged to handle the issues in the middle of the night.

In a typical organization, development teams had very few insights into the world of IT operations, and vice versa. Collaboration between the two departments was minimal, they operated as functional silos. Andrew Clay Shafer would later call this disconnect a "wall of confusion,"[2] referring to the lack of visibility between the two groups as well as the opposing, even conflicting, goals of the two departments. With development teams focused on the speed of building and delivering new features, while operations teams were focused on stability, reliability, and "keeping the lights on" for the revenue-generating systems already running in production, the disconnect was inevitable.

[2] Shafer, A.C. AgileRoots 2009 - Agile Infrastructure. https://www.youtube.com/watch?v=Y_u84PNrX9g